

Spring 2026

Dr. Michael Alfaro | UCLA Ecology & Evolutionary Biology

Your poster tells a **STORY**: Question -> Data -> Analysis -> Answer

[illegible]

	Refs + Acks	
(18x6)		
FOOTER STRIP (methods, fe callouts)		
		36

The poster is 48" wide x 36" tall. Four hero figure panels fill the Results section. The text sections (Intro, Methods, Discussion, Conclusions) **frame** the figures.

The Four Hero Figures

Panel A – Color Gamut

Recipe: color-gamut.qmd

What it shows: 3D RGB convex hull of the clade's color cluster centers

Question it answers: *What colors does your clade have?*

Output: color-gamut-hero.pdf (8"x8")

Panel B – Phylomorphospace

Recipe: phylomorphospace.qmd

What it shows: PC1 x PC2 with fish images at tips and phylogenetic edges

Question it answers: *How is color variation structured in morphospace?*

Output: phylomorphospace-PC1-PC2-hero.pdf (12"x9")

Panel C – Phylogeny + Traits (THE HERO)

Recipe: phylogeny-with-traits.qmd

What it shows: Ladderized tree with PC1 ASR branch colors, tip images, trait columns

Question it answers: *How has color evolved along the tree?*

Output: phylogeny-with-traits-hero.pdf (28"x24")

Panel D – PGLS

Recipe: `pgls-color-traits.qmd`

What it shows: Partial-residual scatters of PC1/PC2 vs trophic level and body size

Question it answers: *Does ecology predict color?*

Output: `pgls-color-traits-hero.pdf` (12"x24")

All Four Hero Figures at a Glance

Chaetodontidae gamut (volume = 0.171, 17.1% of RGB cube)

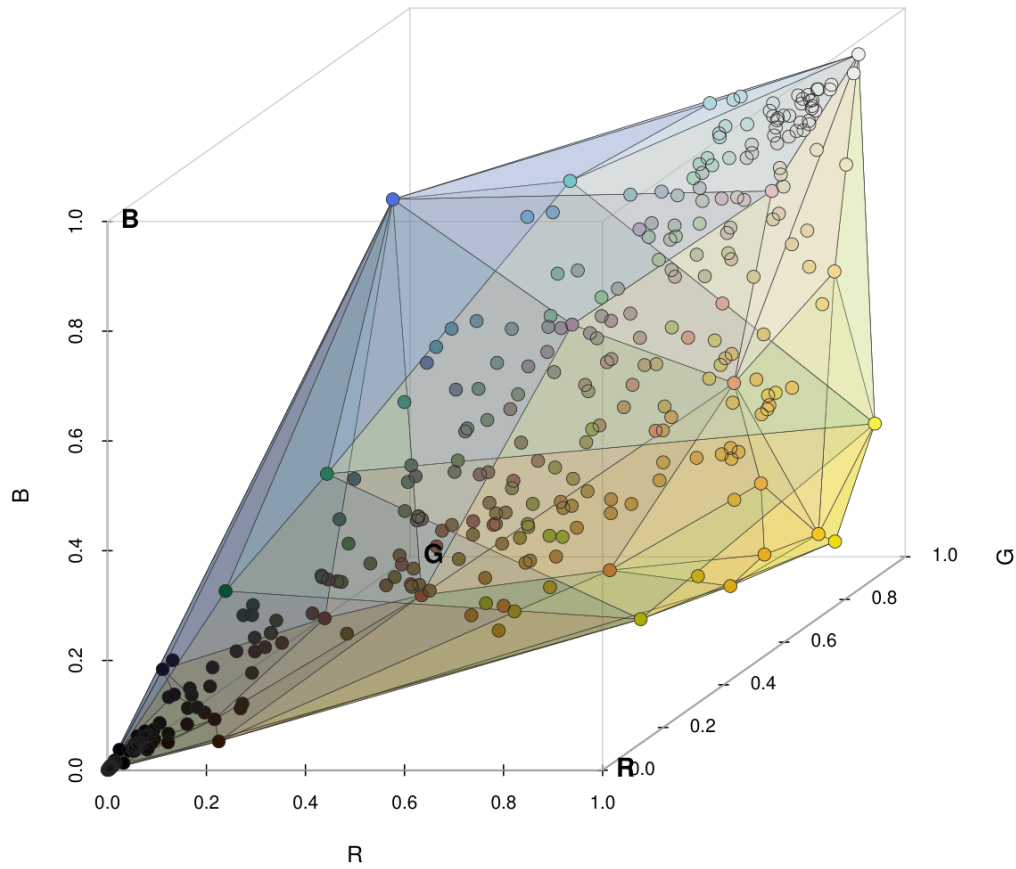


Figure 1: Panel A: Color Gamut

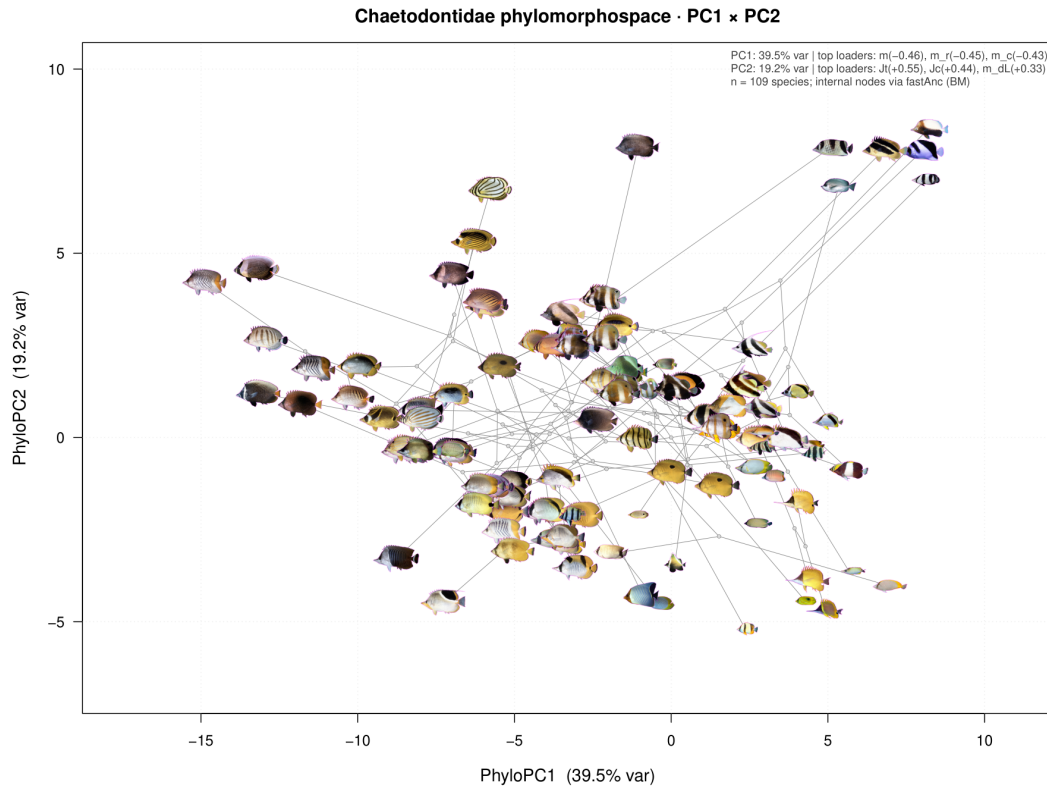


Figure 2: Panel B: Phylomorphospace

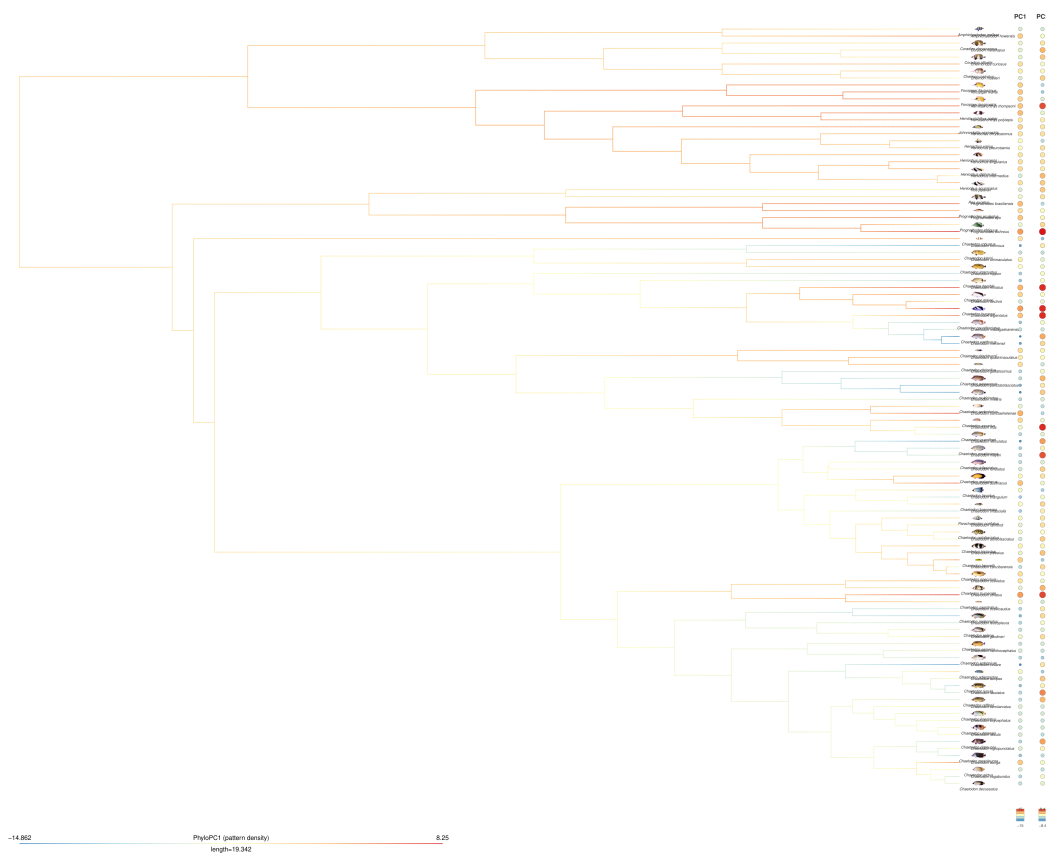


Figure 3: Panel C: Phylogeny + Traits

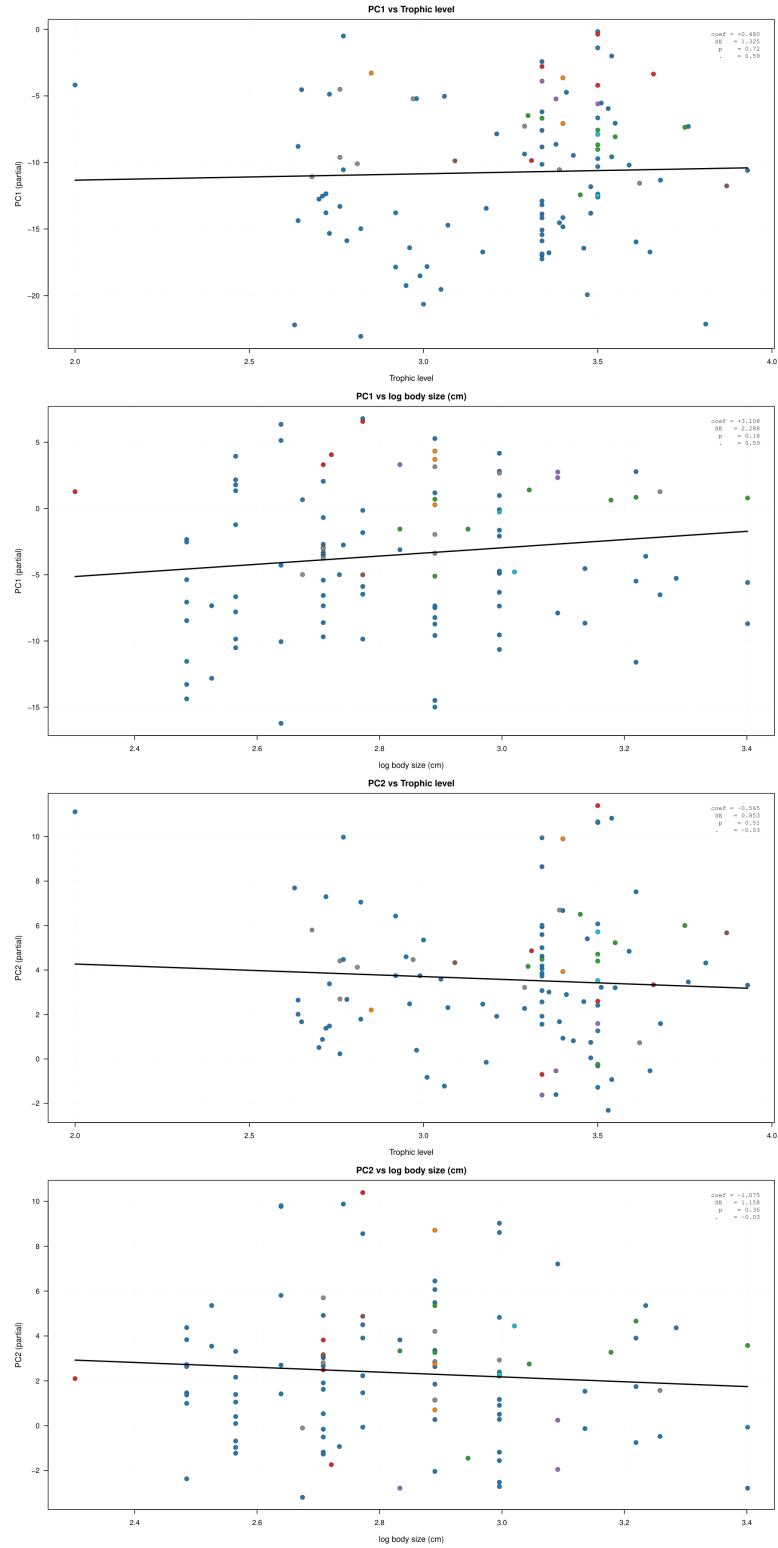


Figure 4: Panel D: PGLS

Workflow Overview

Bundle Export -> pavo classify (cached) -> Run 4 Recipes -> Hero PDFs -> Poster

1. **Bundle export** – Students download their clade’s FishView bundle (HYBRID layout)
2. **pavo classify** – Already cached as `pavo_classify.rds` and `phylo_pca.rds` (saves ~15 min)
3. **Run 4 recipes** – Each recipe ends with a **hero**-PDF chunk. Run it!
4. **Generate hero PDFs** – Four PDFs appear in `figures/`
5. **Drop into poster** – Insert the PDFs into the 48”x36” PowerPoint template
6. **Fill text** – Write Intro, Methods, Discussion, Conclusions around the figures

Each recipe is self-contained. Students do NOT need to understand the full codebase – just run the chunks in order, then grab the hero PDF at the end.

What You Need Before Starting

Files from FishView (already in bundle/):

- `images/` – Exemplar PNGs (one per species, magenta background)
- `<clade>-tree.tre` – Chronogram from Brownstein et al.
- `pavo_metrics.csv` – Full pavo adjacency statistics
- `<clade>_traits.csv` – `body_size_cm`, `trophic_level`, `depth` from FishBase
- `species_k_bg.csv` – `gestalt_k` and `bg_class_id` per species

Cached intermediates (save 15 min per render):

- `pavo_classify.rds` – cached `pavo::classify` output (the slow step)
- `phylo_pca.rds` – cached `phytools::phyl.pca` result

R packages: `pavo`, `ape`, `phytools`, `nlme`, `png`, `geometry`, `scatterplot3d`, `rfishbase`, `tidyverse`

If students see “Error: cannot open file ‘pavo_classify.rds’” – it means the cache was not exported with the bundle. Re-run the classify step (takes ~15 min).

PANEL A: Color Gamut

Recipe: `color-gamut.qmd` – RGB convex hull of pavo cluster centers

Panel A: Color Gamut – Overview

POSTER PANEL A (top-left, 12"x9") – This figure answers: **What is the extent of color diversity in your family?** Look for: where in RGB space does the clade live, and how much of the cube does it fill?

What this recipe does:

1. Loads the cached `pavo::classify` output (cluster centers for each species)
2. Drops the background cluster using `bg_class_id` from the annotation file
3. Pools all surviving cluster centers across the clade into the RGB cube
4. Computes the 3D **convex hull** volume as a fraction of the unit RGB cube
5. Plots a Stoddard-style faceted polyhedron with wireframe reference cube

Key output: A single number (gamut volume as % of RGB cube) plus a 3D scatter

Panel A: Code – Load pavo Data + Extract Cluster Centers

```
## Load packages and the per-species annotation file
library(tidyverse); library(pavo)
library(geometry); library(scatterplot3d)
BUNDLE <- "../..bundle"

ann <- read.csv(file.path(BUNDLE, "species_k_bg.csv"),
                stringsAsFactors = FALSE)
bg   <- setNames(ann$bg_class_id, ann$species_dir)

## Load the CACHED classify output -- students do NOT re-run pavo
classified <- readRDS(file.path(BUNDLE, "pavo_classify.rds"))

## Extract non-background cluster centers as RGB triples
get_palette <- function(rimg_obj, bg_id) {
```

```

rgb_mat <- attr(rimg_obj, "classRGB") # k x 3 matrix
if (is.null(rgb_mat)) return(NULL)
if (!is.na(bg_id) && bg_id >= 1 && bg_id <= nrow(rgb_mat))
  rgb_mat <- rgb_mat[-bg_id, , drop = FALSE] # DROP background
rgb_mat
}

```

This reads the cached `pavo::classify` output. Students do NOT need to re-run `pavo`. Each species contributes $k - 1$ color points (minus the magenta background).

Panel A: Code – Pool Centers + Compute Gamut

```

## Pool all cluster centers across species into one color cloud
palettes <- mapply(
  function(rimg_obj, sp) {
    p <- get_palette(rimg_obj, bg[gsub(" ", "_", sp)])
    if (is.null(p)) return(NULL)
    data.frame(species = sp, R = p[, "R"], G = p[, "G"], B = p[, "B"])
  },
  classified, names(classified), SIMPLIFY = FALSE
)
palettes <- bind_rows(palettes)

## Compute the 3D convex hull of all cluster centers
pts <- as.matrix(palettes[, c("R", "G", "B")])
hull <- convhulln(pts, options = "FA")
vol <- hull$vol
cat(sprintf("Gamut volume: %.4f of unit RGB cube (0.1f%%)\n",
  vol, 100 * vol))

```

What the number means: Volume = 0.0 (all same color) to 1.0 (fills entire RGB cube). Typical reef fish families: 0.05–0.25.

Panel A: What the Output Looks Like

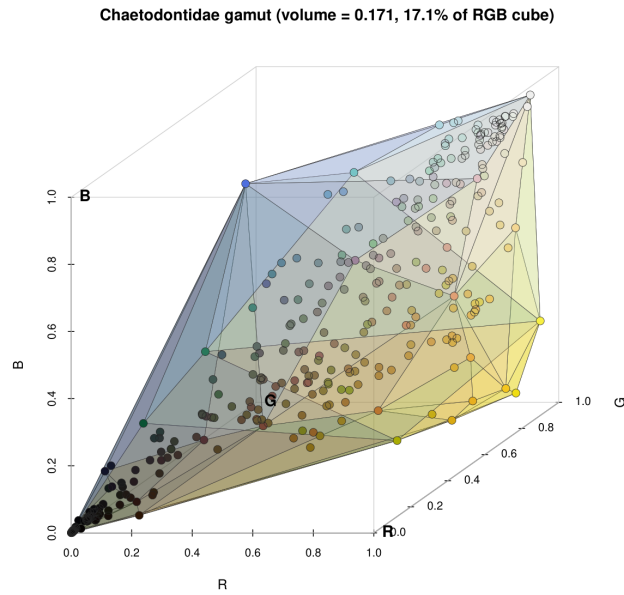


Figure 5: Color Gamut – Stoddard-style 3D RGB scatter

Look for: Where in the cube does the clade live? Is the hull compact (narrow palette) or spread out (chromatic diversity)? For Chaetodontidae, the clade lives mostly in the **yellow-orange-brown** corner with extensions into bright **white** and dark **black**.

Panel A: Rendered In-Document Output

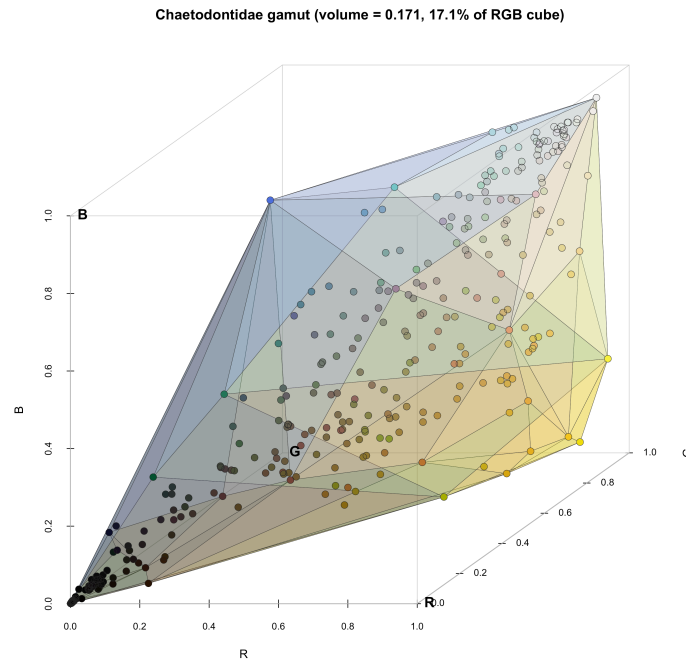


Figure 6: In-recipe figure showing the Stoddard-style gamut with wireframe RGB cube, hull facets, and colored cluster-center points

Wireframe = unit RGB cube (theoretical max). **Semi-transparent facets** = convex hull of your clade's cluster centers. **Colored points** = individual cluster centers, colored by their RGB triple.

Panel A: Hero PDF Export + Caption

```
## Write the poster-ready PDF (8" x 8" for Panel A)
FIGDIR  <- "../figures"
HERO_PDF <- file.path(FIGDIR, "color-gamut-hero.pdf")

pdf(HERO_PDF, width = 8, height = 8)
## (same Stoddard-style plotting code as the in-document figure)
```

```
## ... wireframe cube, hull facets, cluster center points ...
dev.off()
cat(sprintf("Wrote: %s\n", normalizePath(HERO_PDF)))
```

Run this chunk -> get color-gamut-hero.pdf -> drop into Panel A

Caption template: “Figure 1. Color gamut of [YOUR FAMILY] in RGB color space. Each point represents a non-background cluster center from pavo::classify. The semi-transparent polyhedron is the convex hull of the pooled cluster centers (gamut volume = [X.XXX], [X.X]% of the unit RGB cube). Wireframe = theoretical maximum.”

PANEL B: Phylomorphospace

Recipe: phylomorphospace.qmd – PC1 x PC2 with fish images and phylogenetic edges

Panel B: Phylomorphospace – Overview

POSTER PANEL B (top-right, 12”x9”) – This figure answers: **What are the major axes of color-pattern variation, and how are species distributed?** Look for: clusters of similar-looking fish, long edges indicating rapid evolution, and which PC axes drive the separation.

What this recipe does:

1. Loads the cached phylo_pca.rds – phylogenetic PCA of pavo metrics
2. Extracts PC1 and PC2 scores (typically ~60% of total variation)
3. Estimates **internal-node positions** via phytools::fastAnc (Brownian motion)
4. Plots each tip as its **exemplar fish image** (not a dot!)
5. Draws phylogenetic edges connecting tips through reconstructed ancestors

This IS the phylomorphospace – the classic Sidlauskas/Revell visualization.

Panel B: Code – Load PhyloPCA + Reconstruct Ancestors

```

library(ape); library(phytools); library(png)
BUNDLE <- ".././bundle"
IMG_DIR <- file.path(BUNDLE, "images")

## Load cached PCA + tree
pp <- readRDS(file.path(BUNDLE, "phylo_pca.rds"))
ppca <- pp$ppca; tree <- pp$tree
S <- ppca$S
PC1 <- setNames(S[, 1], rownames(S))
PC2 <- setNames(S[, 2], rownames(S))
eig <- diag(ppca$Eval); ve <- 100 * eig / sum(eig)

## Brownian ASR for each axis
anc1 <- fastAnc(tree, PC1)
anc2 <- fastAnc(tree, PC2)

## Combine tip + node coordinates
n_tips <- length(tree$tip.label)
coords <- matrix(NA_real_, nrow = n_tips + tree$Nnode, ncol = 2)
coords[1:n_tips, 1] <- PC1[tree$tip.label]
coords[1:n_tips, 2] <- PC2[tree$tip.label]
coords[(n_tips+1):nrow(coords), 1] <- anc1[as.character((n_tips+1):(n_tips+tree$Nnode))]
coords[(n_tips+1):nrow(coords), 2] <- anc2[as.character((n_tips+1):(n_tips+tree$Nnode))]

```

PC1 and PC2 together capture ~60% of variation in pavo color-pattern metrics. The `fastAnc` call reconstructs where internal nodes sit in morphospace under Brownian motion.

Panel B: Code – Plot with Fish Images

```

## Helper: read PNG and make magenta background transparent
read_fish_png <- function(path) {
  img <- readPNG(path)
  if (length(dim(img)) == 3) {
    if (dim(img)[3] == 3)
      img <- array(c(img, rep(1, prod(dim(img)[1:2]))),
                   dim = c(dim(img)[1:2], 4))
    mag <- img[, , 1] > 0.8 & img[, , 2] < 0.3 & img[, , 3] > 0.8
  }
}

```

```

    img[, , 4][mag] <- 0
  }
  img
}

## Auto-size images based on number of tips
n_species <- length(tree$tip.label)
image_width_in <- max(0.25, min(1.2, 0.6 * sqrt(60 / n_species)))

## Draw phylo edges, then overlay tip images
for (e in seq_len(nrow(tree$edge))) {
  segments(coords[tree$edge[e,1], 1], coords[tree$edge[e,1], 2],
           coords[tree$edge[e,2], 1], coords[tree$edge[e,2], 2],
           col = "grey55", lwd = 0.6)
}
## Then rasterImage() each tip's PNG at its (PC1, PC2) position

```

Each species is plotted as its actual photo, not a dot. Image size auto-scales – 30 tips get big icons, 200 tips get small ones.

Panel B: What the Output Looks Like

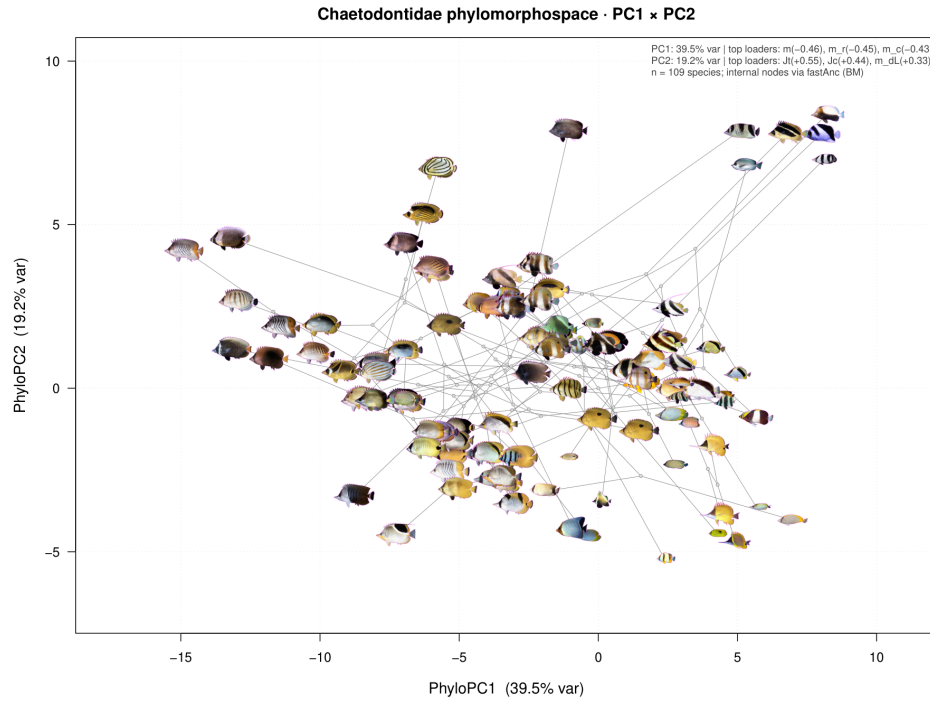


Figure 7: Phylomorphospace – PC1 x PC2 with fish images at tips

Look for: Clusters of similar-looking fish far from the center = convergent evolution on extreme color patterns. Isolated species connected by long edges = evolutionary outliers. Crossing edges are expected (2D projection of a high-dimensional process).

Panel B: In-Recipe Rendered Output

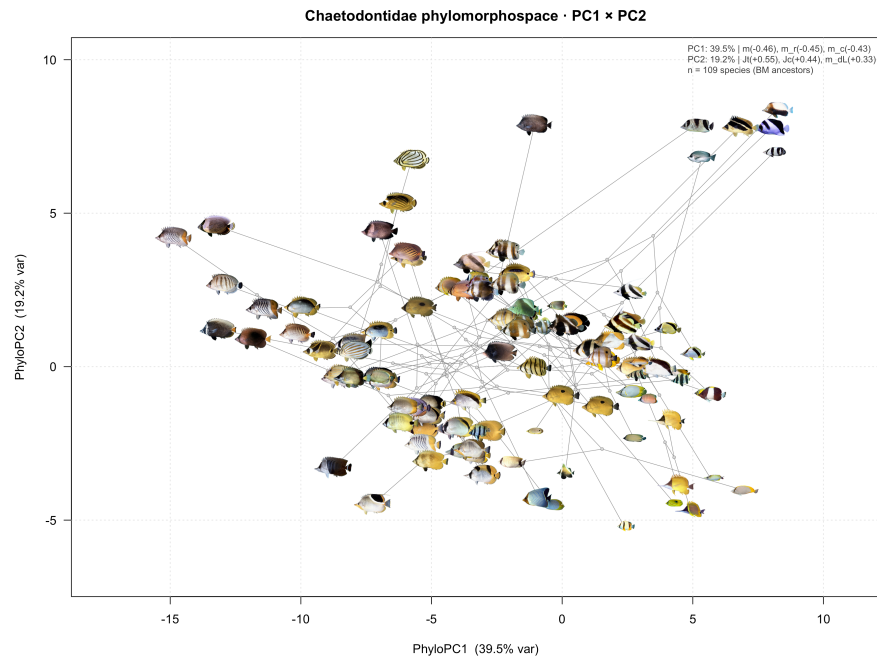


Figure 8: In-document phylomorphospace with variance-explained annotations and top PC loadings in the corner

Tips far from the centroid = extreme color-pattern signatures. **Long edges** = rapid evolution per Ma. **Short edges** = evolutionary stasis. The corner annotation shows variance explained and top loadings per axis.

Panel B: Hero PDF + Caption

```
## The plot_pms() helper wraps the full figure + PDF export
hero_pdf <- file.path(FIGDIR, "phylomorphospace-PC1-PC2-hero.pdf")
plot_pms(hero_pdf, w = 12, h = 9, image_width_in = image_width_in)
```

Run this chunk -> get phylomorphospace-PC1-PC2-hero.pdf -> drop into Panel B

Caption template: “Figure 2. Phylomorphospace of [YOUR FAMILY] color patterns in PC1 x PC2 space (PC1: [X.X]%, PC2: [X.X]% variance). Each tip is plotted as the species’

exemplar image; grey lines connect species through Brownian-motion reconstructed internal nodes (fastAnc). Top PC1 loaders: [list]. Top PC2 loaders: [list]. n = [N] species.”

PANEL C: Phylogeny + Traits

Recipe: phylogeny-with-traits.qmd – THE HERO PANEL (28” x 24”)

Panel C: Phylogeny + Traits – Overview

POSTER PANEL C (center, 28”x24”) – THE BIGGEST PANEL on the poster. This is the visual centerpiece. **Question:** How has color evolved along the tree? Is there phylogenetic signal? **Look for:** branch-color transitions = evolutionary change; trait-circle sorting = phylogenetic signal.

What this recipe builds:

1. Ladderized chronogram with **branches colored by PC1 ASR** (contMap)
2. **Exemplar fish images** at tips (every other tip by default)
3. **Five trait columns** to the right: PC1, PC2, PC3, Trophic level, Body size
4. Each column uses filled circles with color-encoded values

This is the **most information-dense** figure on the poster. Give it the most space.

Panel C: Code – Load Tree + Traits + contMap

```
library(ape); library(phytools); library(png)
BUNDLE <- ".././bundle"

## Load cached PCA and ladderize the tree
pp <- readRDS(file.path(BUNDLE, "phylo_pca.rds"))
tree <- ladderize(pp$tree) # ALWAYS ladderize!
S <- pp$ppca$S
PC1 <- setNames(S[, 1], rownames(S))
```

```

PC2 <- setNames(S[, 2], rownames(S))
PC3 <- setNames(S[, 3], rownames(S))

## Load ecological traits (body_size_cm, trophic_level)
traits <- read.csv(file.path(BUNDLE, "chaet_traits.csv"),
                   stringsAsFactors = FALSE)
keep <- intersect(tree$tip.label, traits$species)
tree <- keep.tip(tree, keep)
traits <- traits[match(tree$tip.label, traits$species), ]

## contMap: continuous trait mapped onto branches
cm1 <- contMap(tree, PC1, plot = FALSE, res = 200)
cm1 <- setMap(cm1, colors = c("#2c7bb6", "#abd9e9", "#ffffbf",
                             "#fdae61", "#d7191c"))

```

Always `ladderize()` your tree! Branch colors: **blue** = low pattern density, **red** = high. Color transitions along a branch = evolutionary change in PC1.

Panel C: Code – Tip Images + Trait Circle Columns

```

## Tip rendering: image + species name at every other tip
show_every_n <- 2 # 1 = every tip, 2 = every other, 3 = sparse
for (i in seq_len(n_tips)) {
  sp <- tree$tip.label[i]
  show_image <- ((i - 1) %% show_every_n == 0)
  if (show_image && file.exists(png_path)) {
    img <- read_fish_png(png_path)
    rasterImage(img, xleft, xright, ybottom, ytop)
  }
  text(..., gsub("_", " ", sp), font = 3) # italic species names
}

## Five trait circle columns to the right of the tree
col_labels <- c("PC1", "PC2", "PC3", "Trophic", "Body (cm)")
col_pals <- list(div_pal, div_pal, div_pal, warm_pal, seq_pal)
for (j in seq_along(col_labels)) {
  for (i in seq_len(n_tips)) {
    enc <- encode(v_all[i], rng[1], rng[2], pal)

```

```

points(xc, tip_y[i], pch = 21,
       bg = enc$col, cex = enc$cex * 1.6)
}
}

```

Tuning show_every_n: < 40 tips = 1 (all images), $40-100 = 2$ (default), $100+ = 3$

Panel C: Code – contMap ASR Branch Colors

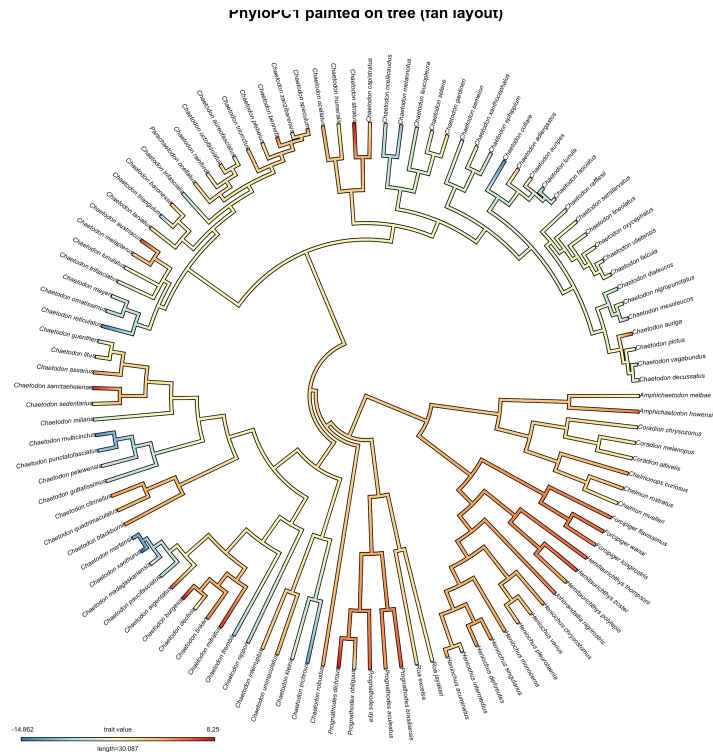


Figure 9: contMap of PC1 showing ASR branch coloring from the phylogenetic-signal recipe

Look for: Clades where branch color shifts dramatically – those are where evolution happened fastest. Uniform-colored clades = evolutionary stasis on PC1. Do nearby species on the tree look similar? (Expected under phylogenetic signal.)

Panel C: What the Output Looks Like

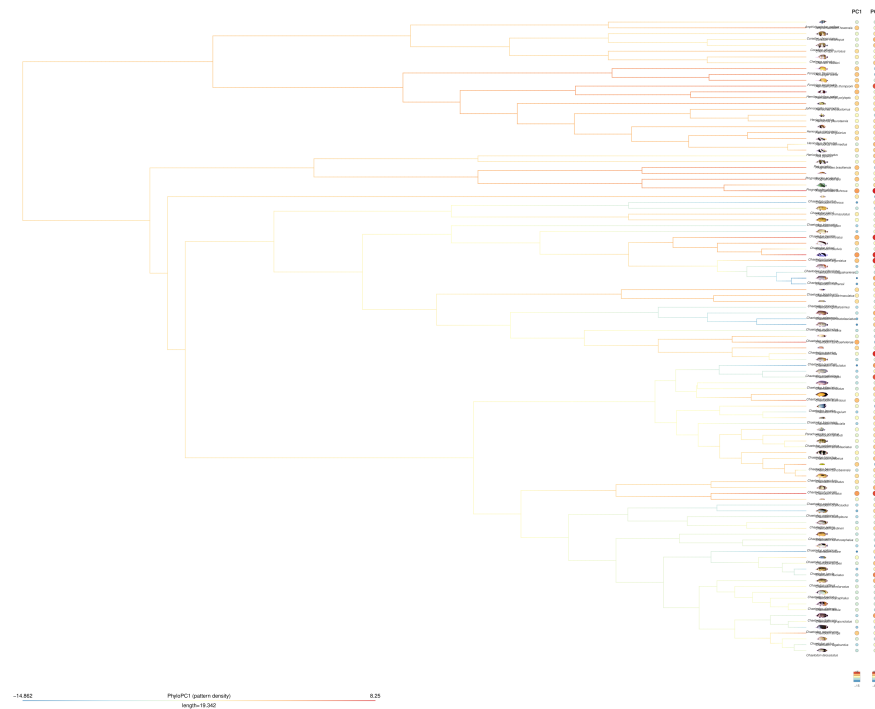


Figure 10: Phylogeny + Traits Hero – the poster centerpiece

Look for: (1) Branch color transitions showing where PC1 evolves. (2) Whether trait circles are sorted by tree position – strong sorting = high K, high lambda. (3) Do species with similar PC1 circles also have similar trophic level?

Panel C: Reading the Figure – TA Coaching Guide

How to guide students through interpretation:

1. **Branch colors:** Where does PC1 shift? Those transitions tell the evolutionary story
2. **Tip images:** Do nearby species on the tree look similar? (Expected under phylogenetic signal)
3. **PC columns:** Are the circles sorted by tree position? (Strong sorting = high K, high lambda)

4. **Trophic + Body size:** Do trait values correlate with PC values across species?

“Compare the trait circles: do species with similar PC1 (same color circle) also have similar trophic level? If yes, ecology and color may be linked – that is what Panel D tests formally.”

Panel C: Hero PDF + Caption

```
## The plot_phylogeny_with_traits() helper wraps the full 28x24 figure
hero_pdf <- file.path(FIGDIR, "phylogeny-with-traits-hero.pdf")
plot_phylogeny_with_traits(hero_pdf, w = 28, h = 24, show_every_n = 2)
```

Run this chunk -> get `phylogeny-with-traits-hero.pdf` -> drop into Panel C (28"x24")

Caption template: “Figure 3. Phylogeny of [YOUR FAMILY] (n = [N] species) with PhyloPC1 ancestral state reconstruction mapped onto branches (blue = low pattern density, red = high). Exemplar images shown at every other tip. Trait columns (right): PC1-PC3 scores (diverging palette), trophic level (warm palette), and body size in cm (sequential blue). Tree from Brownstein et al.; traits from FishBase via rfishbase.”

PANEL D: PGSL

Recipe: `pgls-color-traits.qmd` – Does ecology predict color?

Panel D: PGSL – Overview

POSTER PANEL D (right column, 12"x24") – This figure answers: **After controlling for phylogeny, do trophic level or body size predict color-pattern variation?** Look for: significant slopes ($p < 0.05$), the estimated Pagel's lambda, and whether one genus drives the signal.

What this recipe does:

1. Builds a data frame with PC1, PC2, `trophic_level`, `log(body_size_cm)` for each species

2. Fits PGLS with `nlme::gls` using `corPagel` correlation structure
3. Extracts partial residuals for each predictor
4. Plots a 2x2 grid of partial-residual scatters with stats in the corner

Output: Four scatter panels stacked vertically (PC1 vs trophic, PC1 vs body, PC2 vs trophic, PC2 vs body)

Panel D: What is PGLS? (Background for TAs)

The problem: Related species share traits because of common ancestry. Regular regression treats each species as independent – this inflates sample size and can produce false positives.

The solution: PGLS (Phylogenetic Generalized Least Squares) models the expected covariance between species residuals using the phylogeny.

Pagel's lambda (estimated in the model):

- $\lambda = 0$: Phylogeny does not matter (residuals are independent)
- $\lambda = 1$: Full Brownian motion (related species have strongly correlated residuals)
- λ is estimated from the data – it tells you *how much* phylogeny matters

“Regular regression assumes independence. PGLS accounts for the fact that related species share traits because of common ancestry, not just ecology.”

Panel D: Code – Fitting PGLS Models

```
library(nlme)

fit_pglsls <- function(response_col) {
  f <- as.formula(
    sprintf("%s ~ trophic_level + log_body", response_col))
  ## Try several lambda starting values (optimizer is fragile)
  starts <- c(0.5, 0.9, 0.1, 0.99, 0.0)
  ctrl <- nlme::glsControl(opt = "optim", maxIter = 200,
                           msMaxIter = 200)
  for (lam0 in starts) {
    m <- tryCatch(
```

```

    nlme::gls(f, data = df_use,
              correlation = ape::corPagel(lam0, tree_use,
                                           form = ~species, fixed = FALSE),
              method = "ML", control = ctrl),
    error = function(e) NULL)
  if (!is.null(m)) return(m)
}
## Last resort: pure Brownian (lambda fixed at 1)
nlme::gls(f, data = df_use,
          correlation = ape::corPagel(1, tree_use,
                                       form = ~species, fixed = TRUE),
          method = "ML", control = ctrl)
}
m_pc1 <- fit_pgls("PC1"); m_pc2 <- fit_pgls("PC2")

```

If the model fails to converge, the recipe tries multiple starting values automatically. If ALL fail, it falls back to fixed $\lambda = 1$ (Brownian).

Panel D: Code – Partial Residual Plots

```

## Partial residuals: response minus the contribution of "other"
## predictors, so each scatter shows the ISOLATED effect of the
## focal predictor on the original response scale.
pr_data <- function(m, focal, response_col) {
  b <- coef(m)
  fitted_other <- 0
  for (term in setdiff(names(b), c("(Intercept)", focal)))
    fitted_other <- fitted_other + b[term] * df_use[[term]]
  y_partial <- df_use[[response_col]] - fitted_other
  data.frame(species = df_use$species,
             x = df_use[[focal]], y = y_partial)
}

pr_PC1_trophic <- pr_data(m_pc1, "trophic_level", "PC1")
pr_PC1_body <- pr_data(m_pc1, "log_body", "PC1")
pr_PC2_trophic <- pr_data(m_pc2, "trophic_level", "PC2")
pr_PC2_body <- pr_data(m_pc2, "log_body", "PC2")

```


Each panel: PGLS slope line + coef/SE/p/lambda annotation in corner
Points colored by genus to spot if one genus drives the signal

These show the PARTIAL effect of each predictor – the relationship after removing the other predictor’s effect. The fitted line uses the PGLS slope (not OLS).

Panel D: What the Output Looks Like

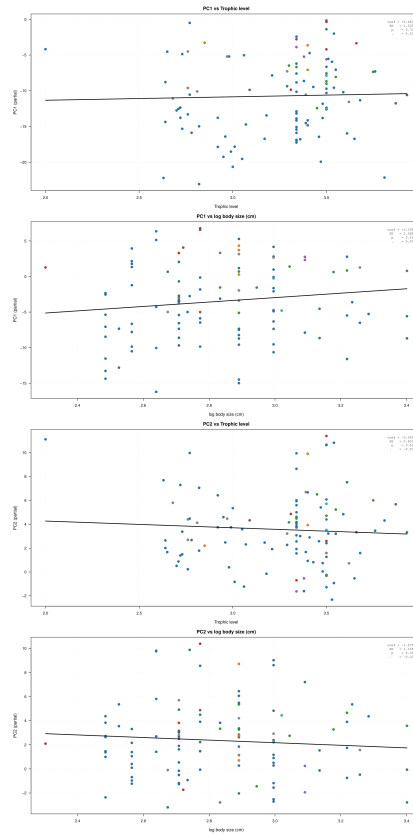


Figure 11: PGLS Hero – 4-panel partial-residual scatter stack

Look for: (1) **Significant slopes** ($p < 0.05$) – the relationship is real after controlling for phylogeny. (2) **Lambda near 1** = strong phylogenetic structure in residuals. (3) **One genus dominating** a cluster = check for Simpson’s paradox.

Panel D: Rendered In-Recipe Output

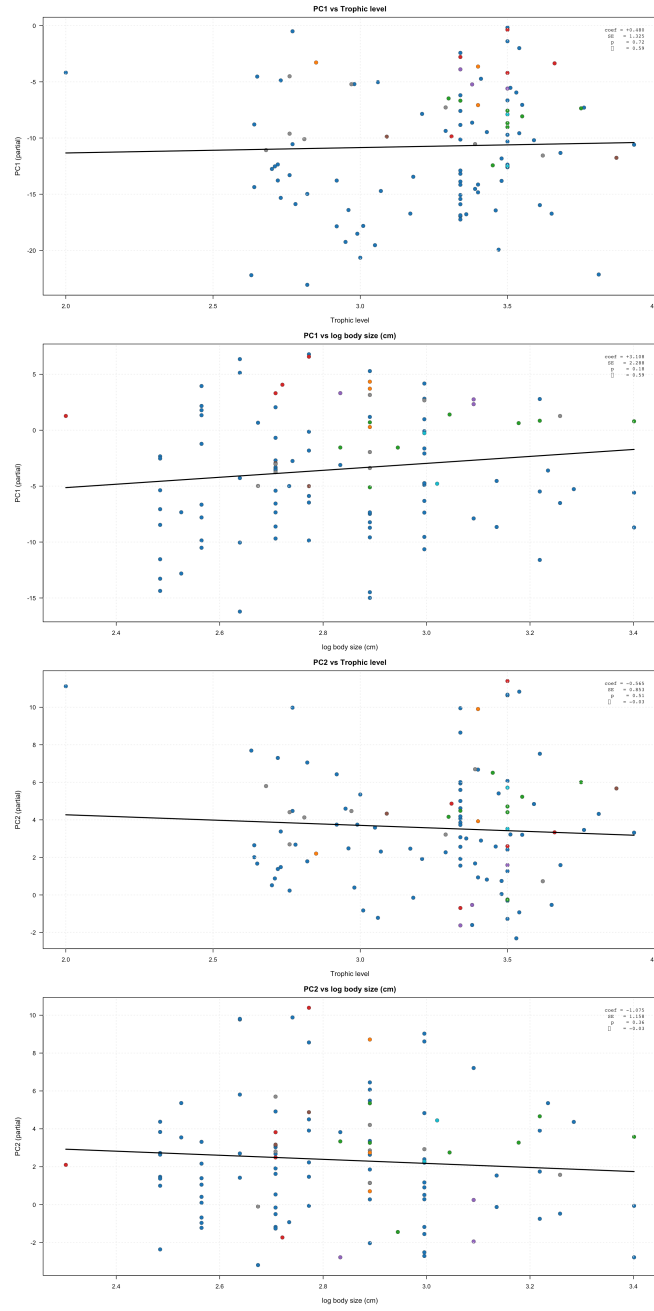


Figure 12: In-recipe PGLS partial-residual plots with genus coloring and corner stats

Each panel shows in the corner: **coef** (PGLS slope), **SE**, **p** (significance), and **lambda** (phylogenetic signal in residuals). Points are colored by genus so you can spot if one genus is driving the signal.

Panel D: Hero PDF + Caption

```
## 4-panel stack: PC1 vs trophic, PC1 vs body, PC2 vs trophic, PC2 vs body
hero_pdf <- file.path(FIGDIR, "pgls-color-traits-hero.pdf")
pdf(hero_pdf, width = 12, height = 24)
layout(matrix(1:4, ncol = 1))
plot_panel(pr_PC1_trophic, "Trophic level", "PC1", m_pc1, "trophic_level")
plot_panel(pr_PC1_body, "log body size", "PC1", m_pc1, "log_body")
plot_panel(pr_PC2_trophic, "Trophic level", "PC2", m_pc2, "trophic_level")
plot_panel(pr_PC2_body, "log body size", "PC2", m_pc2, "log_body")
dev.off()
```

Run this chunk -> get `pgls-color-traits-hero.pdf` -> drop into Panel D

Caption template: “Figure 4. PGLS partial-residual scatters testing associations between color-pattern PCs and ecological predictors in [YOUR FAMILY] ($n = [N]$ species). Each panel shows the partial effect of one predictor after removing the other, fitted with `corPagel` (lambda estimated by ML). Points colored by genus. Slope, p-value, and lambda annotated in each panel.”

If $p < 0.05$, the relationship is significant after controlling for phylogeny. State this clearly in your Discussion section.

Writing the Poster Text

The figures tell the story. The text frames it.

Writing Your Poster Text

Golden rule: A viewer should understand your main finding from the hero figure + conclusion bullets alone (the 6-foot rule). The text sections add context and nuance.

Text budget for a 48”x36” poster:

Section	Word count	Sentences
Introduction	80-120 words	3-4 sentences
Research Question	1-2 sentences	
Hypothesis	2-3 bullets	
Methods	5-6 bullets	
Discussion	80-120 words	3-4 sentences
Conclusions	2-3 bullets	
Future Directions	1-2 bullets	
Acknowledgments	2-3 sentences	
References	5-7 citations	

Introduction Template

Fill in the blanks:

“[YOUR FAMILY] ([common name]) are a family of [N] species distributed across [geographic range]. They exhibit [describe color pattern variation in 1 sentence]. Color patterns in fishes serve multiple ecological functions including [2-3 functions: crypsis, aposematism, mate recognition, species recognition, social signaling]. We used quantitative image analysis (pavo) and phylogenetic comparative methods to investigate [1-sentence version of your question].”

Tips:

- Lead with the biology, not the methods
- Name the family and give species count
- End with your specific question
- 3-4 sentences MAX – this is a poster, not a paper

Research Question + Hypothesis

Research Question: “How are color patterns distributed across the phylogeny of [YOUR FAMILY], and do ecological traits (trophic level, body size) predict color-pattern variation after controlling for shared ancestry?”

Hypothesis: “If color patterns in [YOUR FAMILY] are primarily shaped by [ecological selection / drift / phylogenetic conservatism], then we predict:

1. Phylogenetic signal in color PCs will be [high ($K > 1$) / moderate ($K \sim 1$) / low ($K < 1$)]
2. [Trophic level / body size] will [significantly / not significantly] predict color-pattern variation in PGLS
3. The gamut volume will be [large / small] relative to other reef fish families”

State ONE question. Then state testable predictions. The predictions should connect to specific panels on the poster.

Methods Template

Methods (as a bullet list):

- **Image source:** Standardized exemplar images from FishView ($n = [N]$ species), exported as HYBRID-layout PNGs
- **Color analysis:** `pavo::classify` with $k = [\text{range}]$ clusters per species; cluster centers pooled for gamut computation
- **Phylogenetic PCA:** `phytools::phyl.pca` on $[X]$ `pavo` adjacency metrics under Brownian correlation
- **Phylogenetic signal:** Blomberg’s K and Pagel’s λ for PC1-PC3 (`phytools::phylosig`)
- **PGLS:** `nlme::gls` with `corPagel` correlation structure; PC1 and PC2 regressed on trophic level + $\log(\text{body size})$
- **Tree:** $[N]$ -tip chronogram from Brownstein et al. (2025), pruned to exemplar species

Keep it to 5-6 bullets. The code is in the recipes – the poster just summarizes.

Discussion + Conclusions

Discussion (lead with your main finding):

“Color patterns in [YOUR FAMILY] show [strong/moderate/weak] phylogenetic signal ($K =$ [value] for PC1, $\lambda =$ [value]). The gamut volume of [X.XXX] indicates a [narrow/broad] chromatic range compared to [other families / Alfaro et al. 2019 benchmarks]. [Trophic level / body size] [was / was not] a significant predictor of color-pattern variation after controlling for phylogeny (PGLS: $\text{coef} =$ [value], $p =$ [value], $\lambda =$ [value]). This suggests that [ecological selection / phylogenetic conservatism / neutral evolution] plays a [primary / secondary] role in shaping color patterns in this family.”

Conclusions (2-3 bullets):

- Color patterns in [FAMILY] are [conserved/labile], with [K interpretation]
- [Trait] [does/does not] predict color after controlling for phylogeny ($p =$ [value])
- Compared to butterflyfishes (Alfaro et al. 2019), [FAMILY] shows [more/less/similar] color diversity

References – Every Poster Should Cite These

1. Alfaro, M. E., Karan, E. A., Schwartz, S. T., & Shultz, A. J. (2019). The evolution of color pattern in butterflyfishes (Chaetodontidae). *Integrative and Comparative Biology*, 59(3), 604–615. <https://doi.org/10.1093/icb/icz119> – The benchmark comparison for all reef fish color-pattern analyses in this course.
2. Blomberg, S. P., Garland, T., & Ives, A. R. (2003). Testing for phylogenetic signal in comparative data: behavioral traits are more labile. *Evolution*, 57(4), 717–745. <https://doi.org/10.1111/j.0014-3820.2003.tb00285.x> – K statistic: the standard measure of phylogenetic signal for continuous traits.
3. Revell, L. J. (2012). phytools: an R package for phylogenetic comparative biology (and other things). *Methods in Ecology and Evolution*, 3(2), 217–223. <https://doi.org/10.1111/j.2041-210X.2011.00169.x> – The software behind contMap, phyl.pca, phylosig, fastAnc.
4. Endler, J. A. (2012). A framework for analysing colour pattern geometry: adjacent colours. *Biological Journal of the Linnean Society*, 107(2), 233–253. <https://doi.org/10.1111/j.1095-8312.2012.01937.x> – Theoretical basis for the pavo adjacency metrics.

5. Maia, R., Gruson, H., Endler, J. A., & White, T. E. (2019). pavo 2: New tools for the spectral and spatial analysis of colour in R. *Methods in Ecology and Evolution*, 10(7), 1097–1107. <https://doi.org/10.1111/2041-210X.13174> – The software for image classification and color-pattern quantification.
 6. [Your clade-specific paper] – e.g., the primary systematics or color-ecology paper for your family.
-

Putting It All Together

From hero PDFs to a printed poster

Opening the Template

1. Download `poster-template-48x36.pptx` from the Final Project page on BruinLearn
2. Open in PowerPoint or Keynote
3. You will see **grey placeholder boxes** where the hero figures go

The template has:

- Header strip with title, authors, course info
- Four labeled panel areas (A, B, C, D) with placeholder boxes
- Text areas for Introduction, Methods, Discussion, Conclusions
- Footer strip for references and acknowledgments

Do NOT change the page size. It is set to 48”x36” landscape. If PowerPoint warns about page size, click “OK” to keep it.

Inserting Figures

For each hero PDF:

1. Click on the grey placeholder box for that panel
2. Delete the placeholder
3. **Insert -> Picture -> From File ->** select your hero PDF
4. Resize to fill the panel area. **Hold Shift** to maintain aspect ratio.

Panel	Hero PDF	Size on poster
A (top-left)	color-gamut-hero.pdf	12" x 9"
B (top-right)	phyломorphospace-PC1-PC2-hero.pdf	12" x 9"
C (center)	phylogeny-with-traits-hero.pdf	28" x 24"
D (right column)	pgls-color-traits-hero.pdf	12" x 24"

If PowerPoint cannot import PDF directly, open the PDF in Preview, export as PNG at 300 DPI, then insert the PNG.

Final Checklist

- ☐ All 4 hero figures inserted (Gamut, Phylomorphospace, Phylogeny+Traits, PGLS)
- ☐ All text sections filled:
 - ☐ Title (under 15 words, includes family name + key question)
 - ☐ Authors + affiliations
 - ☐ Introduction (3-4 sentences)
 - ☐ Research Question (1-2 sentences)
 - ☐ Hypothesis (2-3 testable predictions)
 - ☐ Methods (5-6 bullets)
 - ☐ Discussion (3-4 sentences, lead with main finding)
 - ☐ Conclusions (2-3 bullets)
 - ☐ Future Directions (1-2 bullets)
 - ☐ References (5-7 citations)
 - ☐ Acknowledgments (Dickey Collection, TLC Grant, teaching team)
- ☐ Figure captions below each figure
- ☐ Font sizes checked:
 - ☐ Title: 72-96 pt bold
 - ☐ Section headers: 36-48 pt bold

- ☐ Body text: 24-28 pt
- ☐ Figure captions: 18-20 pt
- ☐ References: 16-18 pt
- ☐ Color palette consistent across all figures
- ☐ Exported as PDF for printing (File -> Export -> PDF, page size = 48"x36")

Common Mistakes

Mistake	Fix
Wall of text	Cut to 3-4 sentences per section. Let figures carry the story.
Too many figures	Stick to the 4 hero panels. A K-value table is not a figure.
Unlabeled axes	Every axis gets name + units. "PC1" alone is not enough – say "PC1 (chromatic boundary strength, 68% var)"
Default R styling	Use poster-sized fonts (<code>base_size = 20</code>). Remove gray backgrounds.
Tiny figure text	Re-export with larger <code>cex</code> or <code>base_size</code> . Check at actual poster dimensions.
Buried key result	State your main finding in the title or as a callout in Results.
Inconsistent colors	Same genus = same color in every figure. Pick a palette, stick to it.
Missing captions	Every figure needs a self-contained caption below it.
No hypothesis	State predictions BEFORE the results. Connect each prediction to a panel.
Lambda not reported	Always report Pagel's lambda alongside PGLS p-values.
Wrong page size	Poster must be 48"x36" landscape. Check before exporting to PDF.

Timeline

Date	Milestone
Wed May 28	Analysis done. All 4 recipes run. Research question framed.
Fri May 30	Hero PDFs generated. Text sections drafted.
Sun Jun 1	Full poster draft uploaded to BruinLearn for peer review.
Mon Jun 2	Revise based on peer feedback.
Tue Jun 3	Final poster PDF submitted. Print at SciLi.
Wed Jun 4	Poster presentations (Session 1)
Thu Jun 5	Poster presentations (Session 2)

The BruinLearn draft deadline (Sun Jun 1) is a **HARD** deadline. Posters uploaded after the deadline will not receive peer review feedback.

Going Further

Elective recipes that can replace or supplement Panel D

Want to Go Further?

These **elective recipes** can replace or supplement Panel D if your group has a specific ecological question:

Recipe	What it tests	Could replace
Depth and Color	PGLS of color metrics vs depth	Panel D
Body Size and Complexity	PGLS of pattern density vs body size	Panel D
Vulnerability and Crypsis	PGLS of conspicuousness vs vulnerability index	Panel D

Recipe	What it tests	Could replace
Growth Rate and Color	PGLS of color vs von Bertalanffy K	Panel D
Diet and Conspicuousness	PGLS of conspicuousness vs trophic level	Panel D
Disparity Through Time	DTT analysis with MDI statistic	Panel B
Evolutionary Model Comparison	BM vs OU vs EB model selection	New panel
Discrete Trait Mapping	Ancestral state reconstruction of presence/absence	Panel C variant

Talk to Rosamari or Dr. Alfaro before swapping a panel. The default 4-panel layout works for every clade – alternatives should be motivated by a specific biological question.

If Your Group Wants to Swap Panel D

The default Panel D tests **PC1/PC2 vs trophic level + body size**. But some clades have more interesting ecological questions:

Depth-specialized clades (Prognathodes, deep-water groups):

- Use `depth-and-color.qmd` instead – tests whether deeper species are less conspicuous

Predation-prone clades (small-bodied, exposed reef dwellers):

- Use `vulnerability-crypsis.qmd` – tests whether more vulnerable species are more cryptic

Fast-growing clades (short-lived reef fish):

- Use `growth-rate-color.qmd` – tests whether fast-growing species invest less in color

All alternative PGLS recipes produce a hero PDF that is the same dimensions as the default Panel D (12"x24"). Just swap the PDF in the poster template.

Where to Get Help

Instructor walkthrough (fully rendered with Chaetodontidae data):

- The instructor site shows every recipe with actual output, so you know what your results should look like

Recipe book (student-facing):

- All 32 recipes with code, explanations, and interpretation guides

Office hours:

- Rosamari: [day/time TBD]
- Dr. Alfaro: by appointment

Common error fixes:

- “Error in keep.tip” -> Species names don’t match between your tree and your trait CSV. Check underscores vs spaces.
- “Error in corPagel” -> PGLS optimizer failed. The recipe tries 5 starting values automatically, but if all fail, check for invariant predictors or extreme outliers.
- “Error in readRDS” -> Cache file not found. Re-run the classify or phyl.pca step (takes 10-15 min).
- “Error in rasterImage” -> PNG file missing. Check that `bundle/images/` has the expected species PNGs.

Summary: The Recipe-to-Poster Pipeline

Step	Recipe	Output	Poster Panel
1	color-gamut.qmd	color-gamut-hero.pdf (8x8)	Panel A
2	phylomorphospace.qmd	phylomorphospace-PanA-PanB-hero.pdf (12x9)	Panel B
3	phylogeny-with-traits.qmd	phylogeny-with-traits-PanA-PanB-Co.pdf (28x24)	Panel C
4	ppls-color-traits.qmd	ppls-color-traits-PanA-PanB.pdf (12x24)	Panel D
5	Write text	Intro, Methods, Discussion, Conclusions	Text sections

Step	Recipe	Output	Poster Panel
6	Assemble	Insert PDFs into <code>poster-template-48x36.pptx</code>	Done!

Each recipe is self-contained. Run the chunks in order. Grab the hero PDF at the end. Drop it into the poster. Fill in the text. Export as PDF. Print.

Good luck with your posters!

Questions? Ask Rosamari or Dr. Alfaro.

EEB 187 – Ecology and Evolution of Color – Spring 2026